ITSM and ITOM OnPrem Specialist Enablement Platform and Innovation Suite upgrade

September 2025
Version 2.19

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Document Information

a) Document History

Version	Contributor(s)	Date	Description of Change
1.0	Srikanth Idury	28th December 2022	Details of deployment
1.1	Srikanth Idury Keith Odom	10 February 2023	Dry run version
1.2	Srikanth Idury Keith Odom	3 March 2023	Updates based on dry run feedback
1.3	Srikanth Idury Keith Odom	24 March 2023	IS bootcamp 230327 Minor changes.
1.4	Michael Ashall	12th May 2023	Copied for combined Bootcamp and updated Appendix
1..4	Michael Ashall	9th June 2023	Review formatting and content
2.0	Michael Ashall	11th June 2023	Release GA V2.0
2.1	Michael Ashall	14th June 2023	Copied for updates from prior Bootcamp
2.2	John Gallagher	21st August 2023	Release GA V2.2
2.3	John Gallagher	22nd September 2023	Release GA V2.3
2.4	John Gallagher	2nd February 2024	Release GA V 2.3
2.5	John Gallagher	14th February 2024	Added rollback instructions
2.6	John Gallagher	12th March 2024	Release GA V 2.6
2.7	John Gallagher/Bill Robinson	27th May 2024	Release GA V 2.7
2.11	Bill, John, Mark	29th Jan 2025	Release GA V 2.11

b) Proprietary Notice

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Document Purpose

BMC have created the following hands-on lab guides:

LAB 1 - Helix Platform Services & ITOM - Architecture and Configuration
LAB 2 - Helix Platform Services & ITOM- Environment Validation
LAB 3 - Helix Platform Services & ITOM - Product Installation*
LAB 4 – ITSM Deployment Overview and Environment Prerequisites
LAB 5 – ITSM Deployment Engine Setup and Environment Health Check
LAB 6 – ITSM Product Installation using Pipelines
LAB 7 – ITSM ITOM upgrade

***This Document**

c) Content of each Lab

Lab 1: "**Architecture and Configuration**" In the Architecture document, you will be guided through understanding the key components of the solution, and what roles they play in installation.

Lab 2: "Environment Validation" In this document, you will use the Health Check Tool (HCT) and other methods to validate a customer's environment for installation.

Lab 3: "Platform Installation" In this document, you will use the Helix installer to deploy one or more applications and troubleshoot installation issues.

Lab 4: "Deployment Environment Prerequisites" To get started with the installation, you must understand the end-to-end process for implementing BMC Helix IT Service Management on your system. To set up your system, you must meet the system requirements, download the installation files, perform the installation, complete the system configurations, and then make the system available to the users.

Lab 5: "Deployment Engine Setup and Environment Health Check" This document will help the reader to perform an installation of the ITSM Deployment engine and environmental health check.

Lab 6: "Product Installation using Pipelines" We use Jenkins to deploy Innovation Suite using pipelines. This document will guide you to complete a successful deployment.

Lab 7: "ITSM & ITOM upgrade" In this document you will be learning how to upgrade ITOM and ITSM *

d) Purpose and Intended Audience

The purpose of this Lab Document exercise is to provide enablement of BMC Helix Innovation Suite installation and ITOM.

Although it would be preferential that the reader has some knowledge of using Innovation Suite products, we have tried to provide details including screen shots to aid a person or persons with no prior knowledge of using the product.

The document is not intended as a replacement of the official BMC documentation which is available for a product release. It is solely intended to complement the official documentation with the links to these in Addendum.

Note: BMC Education Courses are available here

e) Objectives

The objective of this document is to help the reader themselves to perform ITSM deployment defined within.

f) Lab Duration

The estimated time for this Lab is approximately 8 hours to configure and complete the upgrade.

Upgrade overview

We will be upgrading Common Services to 25.2 and Innovation Suite to 25.2.01. Please note, 25.1.01 to 25.2.01 deployment mode is UPGRADE not UPDATE – this is a change from earlier releases.

If there aren't any errors the upgrade may take between 6 and 10 hours. The duration may increase depending on how much data is present in the Innovation Suite database. Due to this timeframe, we recommend starting the upgrade pipeline towards the end of your day to allow it to run overnight.

You won't have to do as much configuration as the Upgrade Tool copies over the previous configuration from the "infra.config" as well as adding parameters for the new version. On the Jenkins Deployment Engine side, you will be updating the GIT repositories to use 25.2.01 and reusing many of the same saved parameters from the fresh installation.

Preparation

Always check product compatibility in the system requirements page before upgrading. Ingress Nginx is dependent on the K8s version in use. As we are doing an upgrade most of the infrastructure prerequisites are already in place. For your reference:

https://docs.bmc.com/xwiki/bin/view/IT-Operations-Management/On-Premises-Deployment/BMC-Helix-IT-Operations-Management-Deployment/itomdeploy252/Planning/System-requirements/

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Planning/System-requirements/

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Best-practices-for-upgrade/

a) Installation files

Download the files for the upgrade from Windows Desktop, ***ReadyTech →Inbox*** folder:

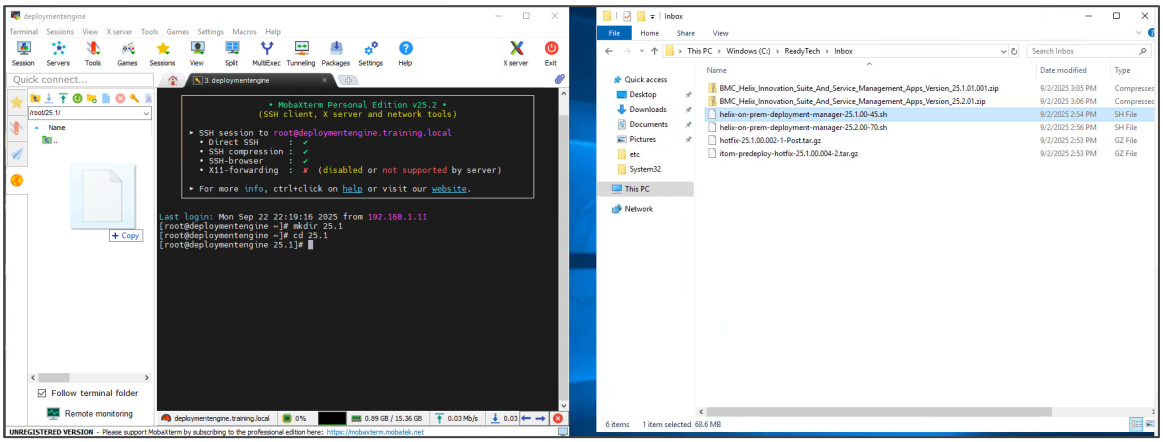
To download a file to the deploymentengine system you can use the WinScp or use the MoboXterm to Drag and Drop the files on the DeploymentEngine.

Steps:

1. Open MoboXterm and connect to deployment engine, credentials are root/password.
2. Create a folder 25.2 on root folder of deploymentengine.
3. Change directory to 25.2.

   # cd 25.2

4. Make sure the Follow Terminal folder is checked.
5. Open the Readytech→Inbox folder on the windows Desktop.
6. Drag *helix-on-prem-deployment-manager-25.2.00-70.sh* to /root/25.2 folder



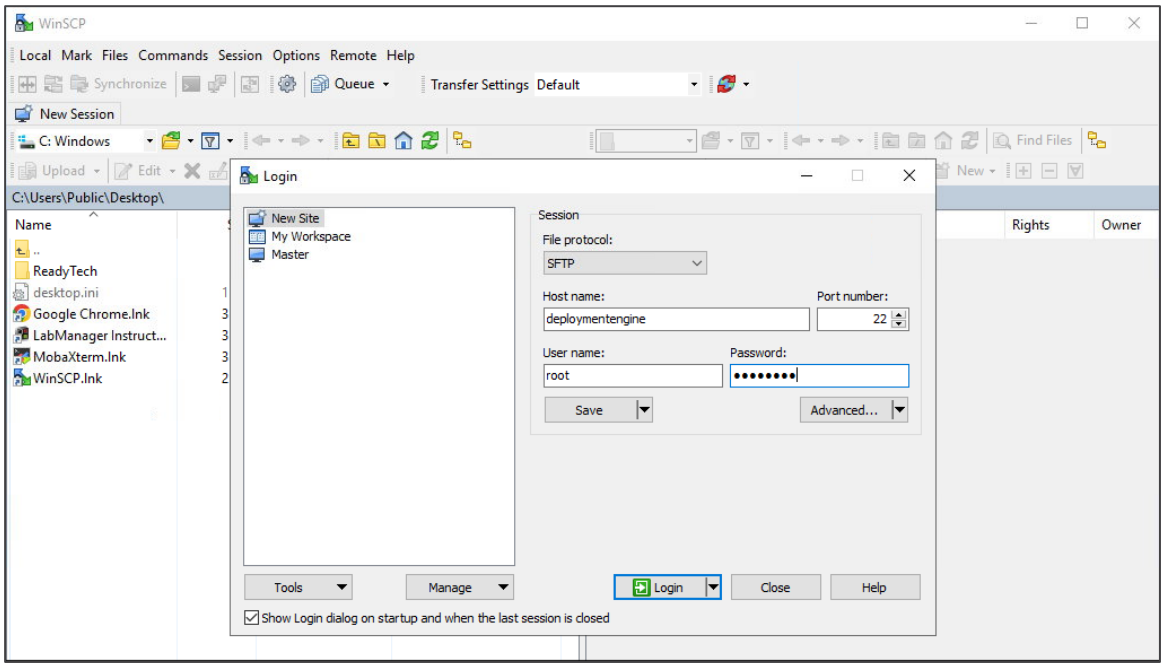
Alternatively, open a session to the deploymentengine system with WinSCP, open the Readytech/Inbox folder navigate to the file you want and drag it deploymentengine /root/25.2 folder.:

1. Open WinSCP from the Windows Dekstop.
2. Click on New Site.
3. Enter below details

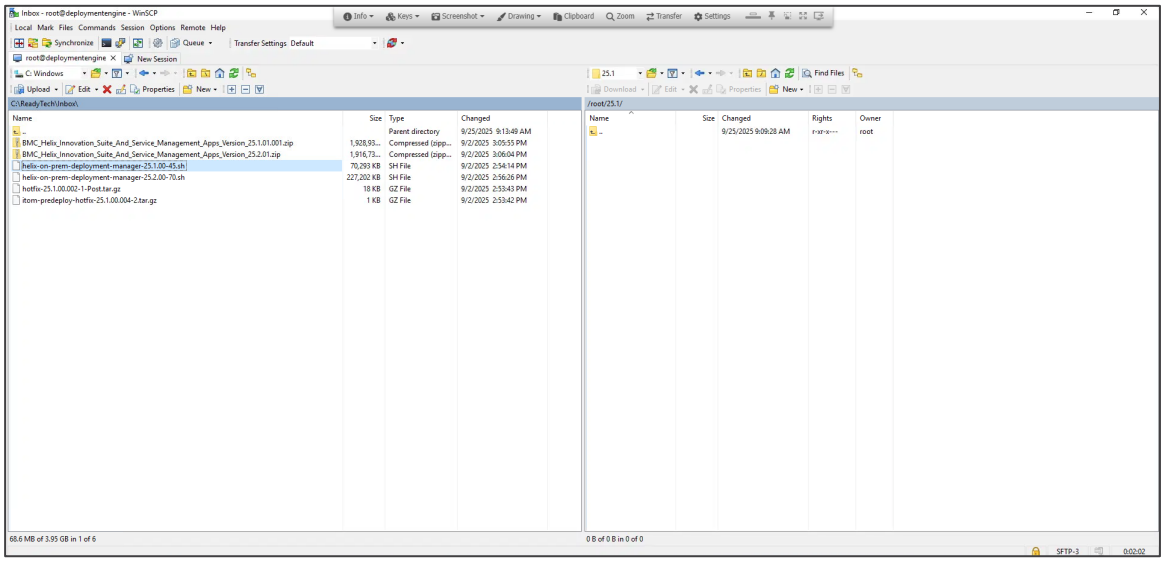
   Hostname: *Deployment Engine*

   Username: *root*

   Password: *password*
4. Click Login and Click Yes on the Warning pop-up.



5. Drag helix-on-prem-deployment-manager-25.1.00-45.sh to /root/25.1 folder



b) GTK3

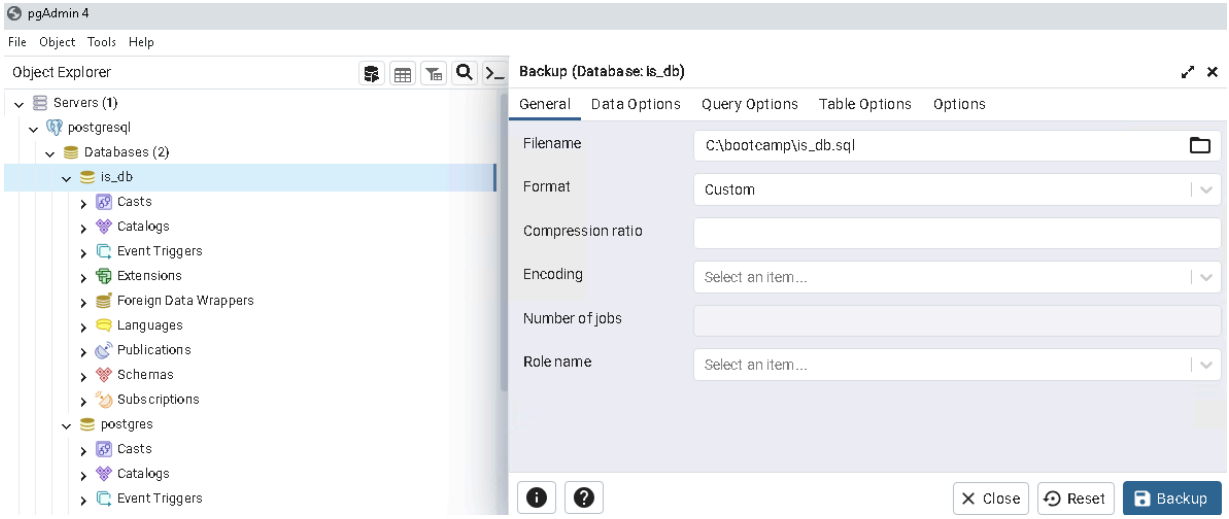
Install the gtk3 package using the following command:

```
yum install -y gtk3
```

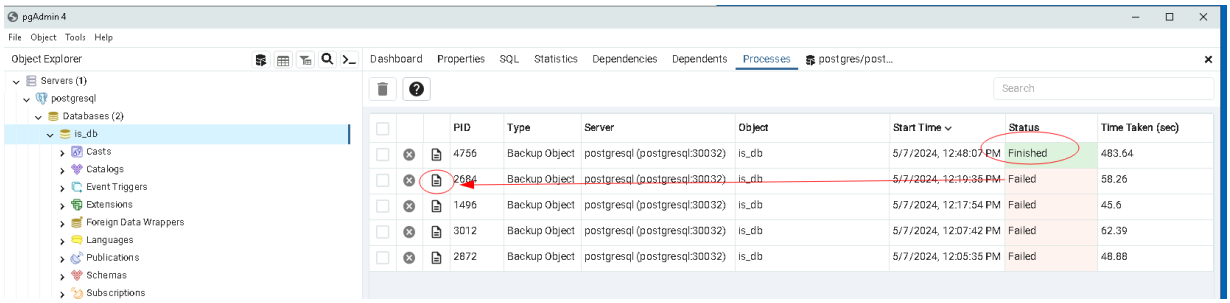
c) Innovation Suite Database

Backup the Innovation Suite Database as follows:

1. Launch PGAdmin from the Windows Jumpbox
2. Right click on is_db, select backup and type “is_db” as the filename and click “Backup”. This process may take 10-15 minutes to complete



Confirm the backup was successful by checking the Processes tab



If the process failed, review the log and resolve the problem before proceeding with the upgrade

d) AR Server License

As detailed in the Lab 6 - 5.11, make sure you have a valid AR Server license before running the upgrade. If the license has expired, check the Bootcamp community site for a newer one.

Expiration Day: 23-Oct-25 License Key: AWU-675772

Prepare for ITOM upgrade

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Platform-Common-Services/Upgrading-BMC-Helix-Platform-Common-Services-version-from-25-1-00-to-25-2-00/

Task: Before you begin

Note: The installer and hotfixes were downloaded in a previous step.

First, we must extract the 25.2 installer files. Hotfixes may be available for any given release, so check the product documentation and/or EPD site before deploying.

a) Extract the 25.2 installer files into a new working directory (if its not done already) distinct from the 25.1 location. For example: create a new working directory called 25.1.00 under root. If you extracted the 25.1 installer files in /root, DO NOT EXTRACT 25.2 in /root

```
$ cd /root
$ mkdir -p /root/25.2.00
$ cp helix-on-prem-deployment-manager-25.2.00-70.sh /root/25.2.00
$ cd /root/25.2.00
$ sh helix-on-prem-deployment-manager-25.2.00-70.sh
```

b) Select Yes (Y) on the License Agreement. This location will be referred to as <extract_dir>

c) Copy the pem certificate to the 25.2.00 certs directory:

```
$ cp /root/class_kit/certs/custom_cacert.pem <extract_dir>/helix-on-prem-deployment-manager/commons/certs/custom_cacert.pem
```

Upgrade utility

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Platform-Common-Services/Upgrading-BMC-Helix-Platform-Common-Services-version-from-25-1-00-to-25-2-00/#HTask1:Toruntheupgradeconfigurationutility

Task 1: To run the upgrade configuration utility

The installer provides a utility to populate the infra.config, deployment.config and secrets.txt with values from the running environment that are required for the upgrade to run. To run the utility change into the "25.2.00/helix-on-prem-deployment/manager/utilities/upgrade" directory and run the utility and supply the common services namespace as an argument:

```
$ cd <extract_dir>/helix-on-prem-deployment-manager/utilities/upgrade
$./upgrade-configuration.sh helixade
```

After the script completes review the output as you may need to set some values for any new parameters that have been added in the new release.

Note: Before migrating data from PostgreSQL database version 15.x to 17.x. in the next task, you must update the new (25.2) deployment.config with the original product selections (Lab 2, 9.6.1 – leave INSTALL_MODE=upgrade)

Postgres Migration

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Platform-Common-Services/Upgrading-BMC-Helix-Platform-Common-Services-version-from-25-1-00-to-25-2-00/#HTask2:TomigratedatafromPostgreSQLdatabaseversion15.xto17.x

Task 2: To migrate data from PostgreSQL database version 15.x to 17.x

1. Navigate to **helix-on-prem-deployment-manager/utilities/migration/postgres** and locate the **patroni-pg-migration.sh** file.
2. To migrate your data from PostgreSQL database 15.x to 17.x, run the following command:
   `./patroni-pg-migration.sh migrate`

Upgrade script

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Platform-Common-Services/Upgrading-BMC-Helix-Platform-Common-Services-version-from-25-1-00-to-25-2-00/#HTask3:ToupgradeBMCHelixPlatformCommonServices

Task 3: To upgrade BMC Helix Platform Common Services

a) Ensure DEPLOYMENT_SIZE is set to **compact** in the "/root/25.2.00/helix-on-prem-deployment-manager/configs/deployment.config" file

b) As we are using BHOM, leave **Victoria Metrics** set to **Yes** (NOT as per BMC ITSM doc)

c) Ensure that the other settings in the deployment.config are set to what's in the pre-hotfix 25.1 deployment.config. The upgrade should take about an hour, if it completes in a few minutes then it didn't really run and you need to check the settings in the deployment.config

d) From the "/root/25.2.00/helix-on-prem-deployment-manager" directory, run the deployment manager script to upgrade BMC Helix Platform Common services by using the following command:

```
$./deployment-manager.sh
```

To monitor for Kubernetes warning events, type the following:

```
$ kubectl get events -n helixade --field-selector type=Warning -w
```

In parallel use K9s to monitor the pod status:

```
k9s -n helixade
```

A successful installation will report the following message:

```
!!!!!!!!!!!!Completed Helix On-prem Upgrade!!!!!!!!!!!!!!!
```

If the installation fails, check the logs located in "/root/25.1.00/helix-on-prem-deployment-manager/logs/deployment.log".

Deployment Engine config

Note: You do **NOT** need to redeploy Jenkins for upgrades or patching.
Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Preparing-for-upgrade/

Task 3: To update the Git repositories
Note that we are using the
BMC_Helix_Innovation_Suite_And_Service_Management_Apps_Version_25.2.01.zip file.

To rename repos with a _25101 extension using one command, you can use a "for" loop. For example, on the command line paste the following text from the "/home/git/git_repo" directory:

```
for item in CUSTOMER_CONFIGS DB_REFRESH_REPO HELM_REPO ITSM_REPO
PLAYBOOKS_REPO SMARTAPPS_HELM_REPO; do
    mv "$item" "${item}_25101"
done
```

Unzip the **BMC_Remedy_Deployment_Manager_Configuration_Release_25.2.01.zip** file in /home/git/git_repos directory and unzip it.

```
$unzip
BMC_Helix_Innovation_Suite_And_Service_Management_Apps_Version_25.2.01.zip

$unzip BMC_Remedy_Deployment_Manager_Configuration_Release_25.2.01.zip
```

Deployment Pipelines

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Preparing-for-upgrade/#HTask4:Toadddeploymentpipelines

Task 4: To add deployment pipelines
If any of the pipelines or credentials the instructions say to create already exist, you do not need to create them. Depending on the starting version of IS, they may or may not be present.

Innovation Suite Platform and application upgrade

Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Service-Management-to-25-2-01/

Before You Begin: To Add Node configuration

Task 1: To update the deployment pipelines

Task 2: To run the HELIX_ONPREM_DEPLOYMENT pipeline

Many of the parameters values (including passwords) from the previous installation will have been saved. Please check the following sections and update accordingly.

Note: Jenkins has a timeout of 30 minutes so we encourage you save your work by entering a few parameters at a time and clicking the “Rebuild” button with no Products selected.

1.1.1 ENVIRONMENTAL DETAILS

Parameter	BDC LAB environment values
DEPLOYMENT_MODE	UPGRADE
CACERTS_FILE	Upload the same cert as before
SOURCE_VERSION	2025101.1.00.00
PLATFORM_HELM_VERSION	2025201.1.00.00
SMARTREPORTING_SOURCE_VERSION	NA
SMARTAPPS_HELM_VERSION	2025201.1.00.00

1.1.1 PIPELINES

Parameter	BDC LAB environment values
HELIX_GENERATE_CONFIG	Checked
HELIX_PLATFORM_DEPLOY	Unchecked
HELIX_SMARTREPORTING_DEPLOY	Unchecked
HELIX_NONPLATFORM_DEPLOY	Unchecked
HELIX_CONFIGURE_ITSM	Unchecked
HELIX_SMARTAPPS_DEPLOY	Unchecked
HELIX_INTEROPS_DEPLOY	Unchecked
HELIX_FULL_STACK_UPGRADE	Checked
SUPPORT_ASSISTANT_TOOL	Unchecked
HELIX_POST_DEPLOY_CONFIG	Unchecked
HELIX_DR	Unchecked
SCALE_DOWN	Unchecked
HELIX_RESTART	Unchecked

1.1.2 Database, Registry, Storage, Passwords and Secrets

Parameter	BDC LAB environment values
DATABASE_RESTORE	Unchecked
FTS_ELASTICSEARCH_HOSTNAME	opensearch-logs-data.helixade
FTS_ELASTICSEARCH_PORT	9200
FTS_ELASTICSEARCH_USER_PASSWORD	Es_L0g#p@SS
FTS_ELASTICSEARCH_SECURE	Checked

1.1.1 Innovation Suite upgrade

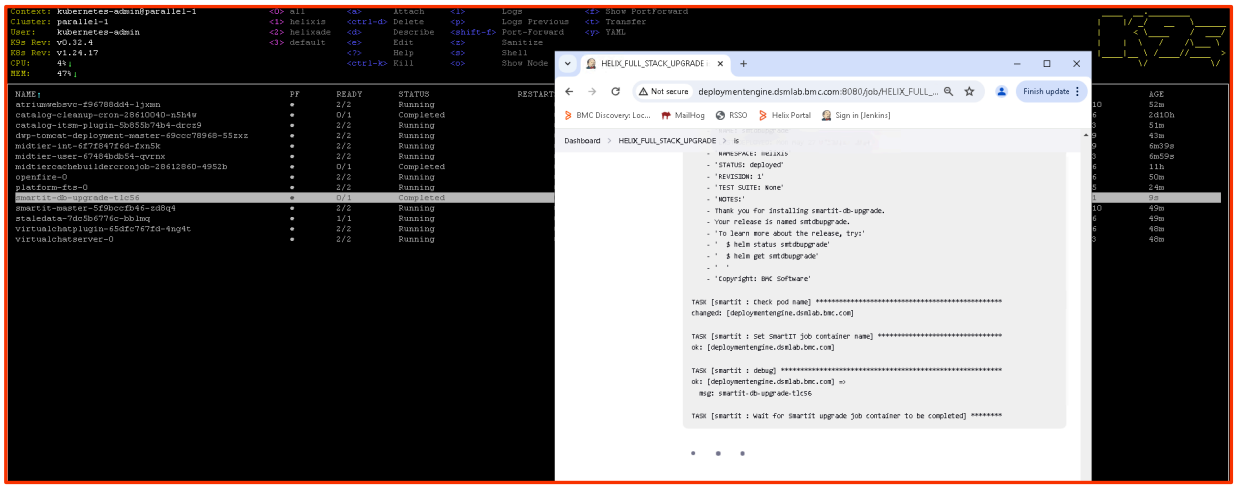
Doc reference: https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25201/Upgrading/Upgrading-BMC-Helix-Service-Management-to-22-1-06/#HTask2:ToruntheHELIX_ONPREM_DEPLOYMENTpipeline
Task: 2

Once you click the rebuild button please proceed to the next section which explains how to monitor and troubleshoot an upgrade.

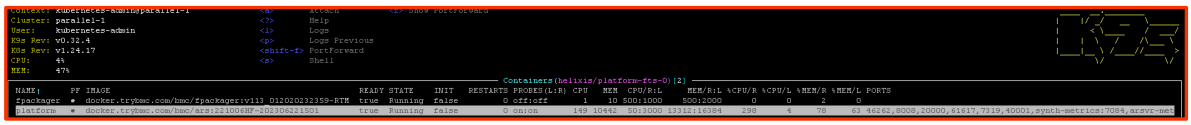
Please note: The upgrade can take over **8 hours** to complete. If the upgrade gets stuck it will present you with a Rollback and Abort hyperlinks in the console log, please click the **Abort** button, do **NOT** select the Rollback link.

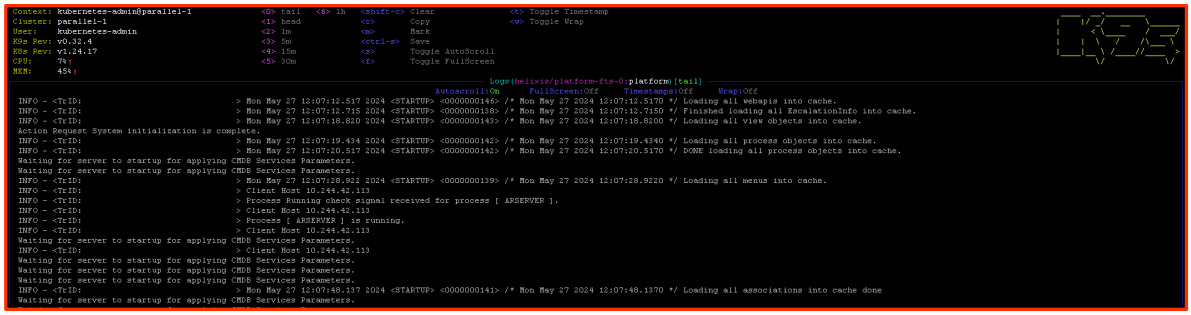
Troubleshooting

1. Monitor the pipeline console log & helixis namespace side by side. For example:



2. If the console log appears to be stuck on a particular pod, check the relevant container log for a more in-depth view of what's going on (select "l" while highlighting the container in K9s) You can also stream container logs via command line. For example:
   `kubectl logs -f <podname> -c <containername> -n helixis`





Note: The platformd2pupgrade pod can take up to an hour to complete but you can see what it's attempting to deploy by checking the container logs or by logging into the shell (ARSystem/db directory) and viewing the relevant D2P log (tail -f ard2pdeploymentactivity.log)

```
Context: kubernetes-admin@parallel-1          <0> tail   <6> 1h   <shift-c> Clear          <t> Toggle Timestamp
Cluster: parallel-1                           <1> head             <c>       Copy           <w> Toggle Wrap
User:    kubernetes-admin                     <2> 1m               <m>       Mark
K9s Rev: v0.32.4                              <3> 5m               <ctrl-s>  Save
K8s Rev: v1.24.17                             <4> 15m              <s>       Toggle AutoScroll
CPU:     5%                                   <5> 30m              <f>       Toggle FullScreen
MEM:     51%
------------------------------------------------ Logs(helixis/platformd2pupgrade-wd8rf:platformd2pupgrade)[tail] ------------------------------------------------
                                              Autoscroll:On      FullScreen:Off     Timestamps:Off     Wrap:Off
 INFO - <TrID:                                > Input values are arcmdb.zip , DB Only AR CMDB 23.3.00 V1 , ars-cmdb_DBONLY_23300_2401201254
 INFO - <TrID:                                > D2P : Package Details : zipfileName : arcmdb.zip packagename : DB Only AR CMDB 23.3.00 V1 and packageversio
 INFO - <TrID:                                > D2P : Importing the package arcmdb.zip
 INFO - <TrID:                                > getPackageStatus : No entries
 INFO - <TrID:                                > isPackageAlreadyDeployed false
 INFO - <TrID:                                > getPackageStatus : No entries
 INFO - <TrID:                                >
  Command for operation Import is "/opt/java/bin/java" -Dlog4j.configurationFile=log4j_d2pcli.xml -cp "/opt/bmc/ARSystem/artools/lib/seqDeplUtilityRemote.
 INFO - <TrID:                                > Waiting for Import operation to get completed.
 INFO - <TrID:                                > Package Imported Successfully.
 INFO - <TrID:                                > Total Elapsed Time : 0 minutes, 18 seconds
 INFO - <TrID:                                > Import operation exited with code 0
 INFO - <TrID:                                > D2P : Checking the ImportStatus
 INFO - <TrID:                                >
  Command for operation ImportStatus is "/opt/java/bin/java" -Dlog4j.configurationFile=log4j_d2pcli.xml -cp "/opt/bmc/ARSystem/artools/lib/seqDeplUtilityR
 INFO - <TrID:                                > Package Name:DB Only AR CMDB 23.3.00 V1
 INFO - <TrID:                                > Version: ars-cmdb_DBONLY_23300_2401201254
 INFO - <TrID:                                > Status: Ready to Deploy
 INFO - <TrID:                                > ImportStatus operation is successful as status is  Ready to Deploy
 INFO - <TrID:                                > D2P :The status for package name  DB Only AR CMDB 23.3.00 V1 and package version is Ready to Deploy
 INFO - <TrID:                                > D2P : Deploying the package with packageName DB Only AR CMDB 23.3.00 V1 and packageVersion ars-cmdb_DBONLY_
 INFO - <TrID:                                > D2P : Performing Operation Deploy as false is set to false.
 INFO - <TrID:                                > getPackageStatus returned 300
 INFO - <TrID:                                >
  Command for operation Deploy is "/opt/java/bin/java" -Dlog4j.configurationFile=log4j_d2pcli.xml -cp "/opt/bmc/ARSystem/artools/lib/seqDeplUtilityRemote.
```

3. Check activity on the kubernetes cluster while performing the deployment:

```
kubectl get events -n helixis --field-selector type=Warning
```

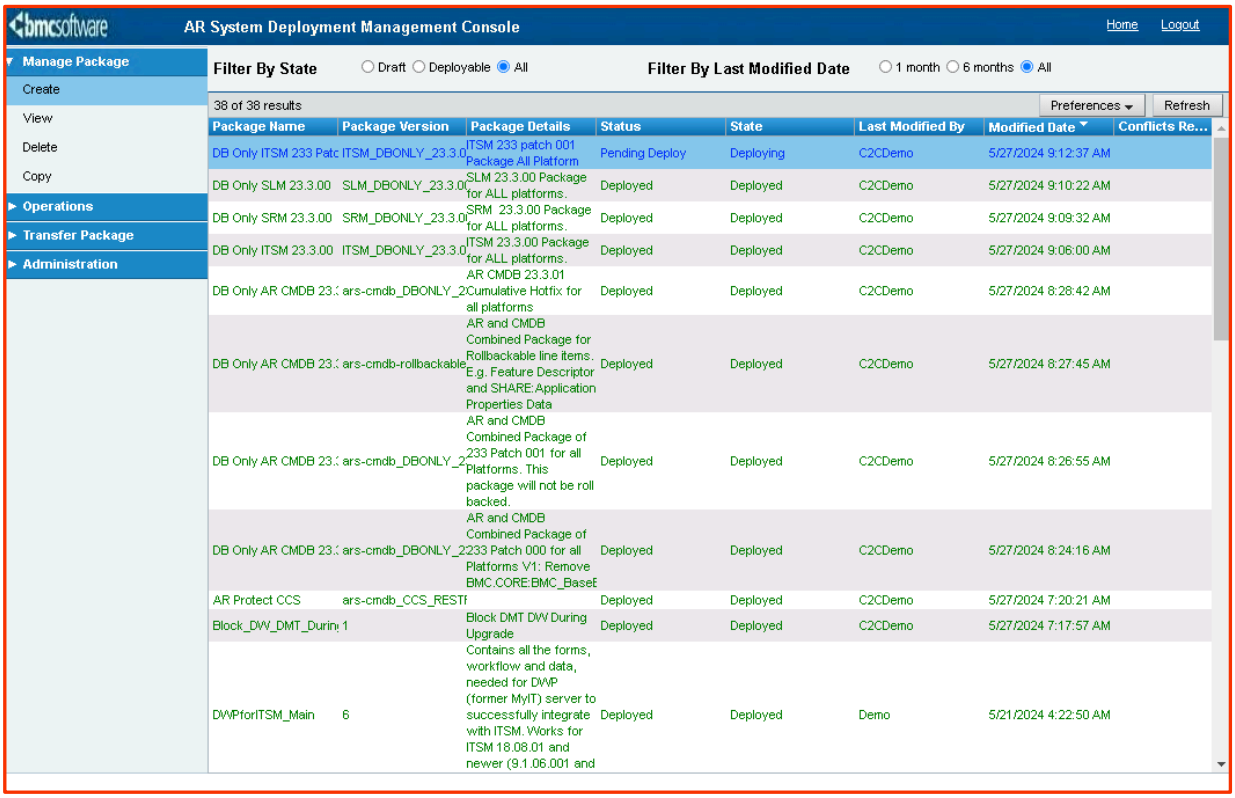
4. Log into the platform-upgrade-0 pod when it's up (select "s" for shell in K9s) and check logs with the most recent timestamp from the "ARSystem/db" directory.

```
<<K9s-Shell>> Pod: helixis/platform-upgrade-0 | Container: platform
platform-upgrade-0:~$ cd ARSystem/db
platform-upgrade-0:~/ARSystem/db$
```

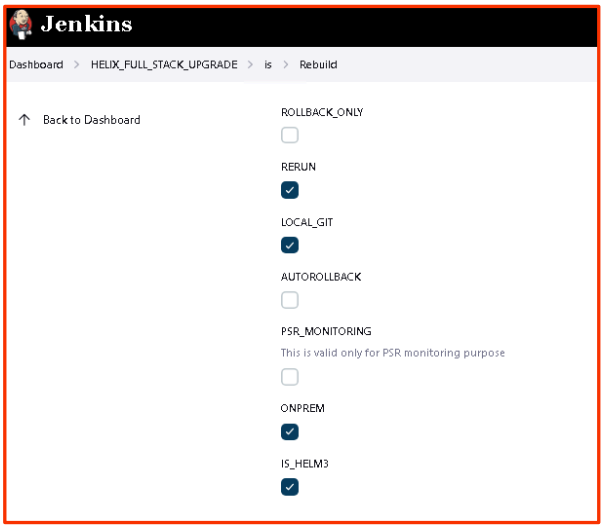
5. Check upgrade progress with helm (although it takes 5 + hours to upgrade you some apps will show as 25.2.01 after an hour)

```
helm ls -n <is-namespace>
```

6. Check the AR System Deployment Management Console for package status



7. If the “HELIX_FULL_STACK_UPGRADE” pipeline fails due to a resource or known config error, rebuild the last failed run and tick the RERUN box. This will resume the upgrade from where it last failed.



To revert Innovation Suite to source version

The following procedure can be used for an unrecoverable failure.

1. Only proceed to step #2 if any of the helm releases have been updated to the version you are trying to upgrade to

```
helm -n helixis ls
```

2. Take note of all active statefulsets & deployments and scale down to zero, for example:

```
kubectl -n helixis get sts,deploy
```

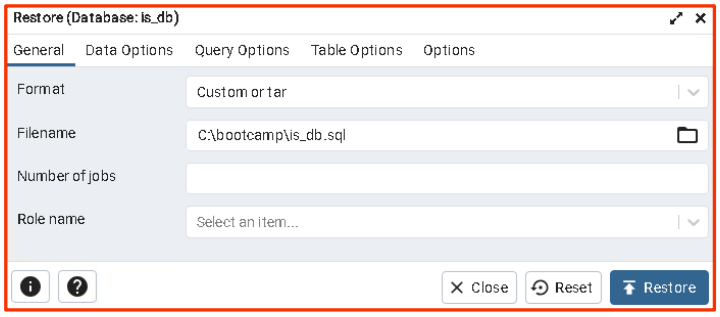
```
[root@deploymentengine ~]# kubectl -n helixis get sts,deploy
NAME                                  READY   AGE
statefulset.apps/openfire             1/1     2d8h
statefulset.apps/platform-fts         1/1     2d8h
statefulset.apps/platform-int         0/0     2d8h
statefulset.apps/platform-sr          0/0     2d8h
statefulset.apps/platform-user        0/0     2d8h
statefulset.apps/virtualchatserver    1/1     2d8h

NAME                                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/assisttool-dep                2/2     2            2           2d6h
deployment.apps/atriumwebsvc                  1/1     1            1           2d8h
deployment.apps/catalog-itsm-plugin           1/1     1            1           2d6h
deployment.apps/clamav                        1/1     1            1           2d8h
deployment.apps/dwp-tomcat-deployment-master  1/1     1            1           2d6h
deployment.apps/dwp-tomcat-deployment-slave   0/0     0            0           2d6h
deployment.apps/midtier-int                   1/1     1            1           2d8h
deployment.apps/midtier-user                  1/1     1            1           2d8h
deployment.apps/smartit-master                1/1     1            1           2d8h
deployment.apps/smartit-slave                 0/0     0            0           2d8h
deployment.apps/staledata                     1/1     1            1           2d8h
deployment.apps/virtualchatplugin             1/1     1            1           2d8h
```

Note: 1/1 or 2/2 indicate the number of pods in an active statefulset or deployment.

```
kubectl scale --all --replicas=0 --namespace=helixis deployment
kubectl scale --all --replicas=0 --namespace=helixis sts
```

3. From the pgAdmin tool, force drop the “is_db” database then right click and create a new database with same name (“is_db”) and restore the DB backup (pre-upgrade) which should take approx. 15 minutes



4. Get helm history:

```
helm -n helixis ls
```

5. Get revision history for relevant release

```
helm -n helixis history platform-fts-helixis
```

6. Rollback to previous working revision (where revision number is date of known last working version):

```
helm -n helixis rollback platform-fts-helixis 1
```

Repeat until all releases below are previous version, for example:

```
[root@deploymentengine ~]# helm -n helixis ls
NAME                          NAMESPACE       REVISION        UPDATED                                   STATUS          CHART
assisttool                    helixis         1               2024-05-21 04:30:49.271721617 -0500 CDT deployed        assisttool-1.0.13
atriumwebsvc-helixis          helixis         3               2024-05-23 11:31:38.616502422 -0500 CDT deployed        atriumwebsvc-22.1.06
catalog-cleanup-cron          helixis         1               2024-05-23 08:04:38.857510202 -0500 CDT deployed        catalog-cleanup-cron-23.3.01
catalog-itsm-plugin-helixis   helixis         3               2024-05-23 11:32:07.13047781 -0500 CDT  deployed        catalog-itsm-plugin-22.1.06
clamav-helixis                helixis         1               2024-05-21 02:00:41.786310979 -0500 CDT deployed        clamav-22.1.06
dwp-tomcat-helixis            helixis         3               2024-05-23 11:32:38.781840003 -0500 CDT deployed        dwp-22.1.06
midtier-int-helixis           helixis         3               2024-05-23 11:33:11.289548277 -0500 CDT deployed        midtier-22.1.06
midtier-user-helixis          helixis         3               2024-05-23 11:33:30.47639671 -0500 CDT  deployed        midtier-22.1.06
midtiercachebuildercronjob    helixis         1               2024-05-23 08:26:59.712720871 -0500 CDT deployed        midtiercachebuildercronjob-23.3.01
openfire-helixis              helixis         2               2024-05-23 08:40:11.5830382 -0500 CDT   deployed        openfire-4.6.3
platform-admin-helixis        helixis         3               2024-05-23 11:35:15.539296904 -0500 CDT deployed        platform-22.1.06
platform-fts-helixis          helixis         4               2024-05-23 11:35:35.287798925 -0500 CDT deployed        platform-22.1.06
platform-int-helixis          helixis         3               2024-05-23 11:35:46.6526811 -0500 CDT   deployed        platform-22.1.06
platform-sr-helixis           helixis         4               2024-05-23 11:35:58.946031761 -0500 CDT deployed        platform-22.1.06
platform-user-helixis         helixis         3               2024-05-23 11:36:09.958637387 -0500 CDT deployed        platform-22.1.06
smartit-master-helixis        helixis         3               2024-05-23 11:36:22.193509744 -0500 CDT deployed        smartit-22.1.06
smartit-slave-helixis         helixis         3               2024-05-23 11:36:33.809878378 -0500 CDT deployed        smartit-22.1.06
staledata-helixis             helixis         3               2024-05-23 11:37:01.498462354 -0500 CDT deployed        staledata-22.1.06
virtualchatplugin-helixis     helixis         3               2024-05-23 11:37:29.974309721 -0500 CDT deployed        virtualchatplugin-22.1.06
virtualchatserver-helixis     helixis         3               2024-05-23 11:37:50.43467022 -0500 CDT  deployed        virtualchatserver-22.1.06
```

Note: Some releases such as the support assistant tool may not change.

7. The pods should all automatically scale up after rolling back with helm, if not scale up as follows:

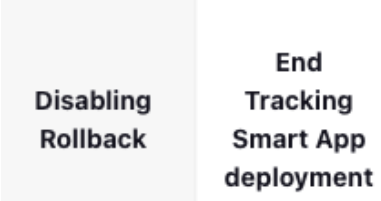
```
kubectl -n helixis scale deploy atriumwebsvc catalog-itsm-plugin dwp-tomcat-deployment-master midtier-int midtier-user smartit-master staledata virtualchatplugin --replicas=1

kubectl -n helixis scale sts openfire platform-fts --replicas=1
```

8. Make sure all pods are in a running state (1/1 or 2/2).
9. Check to make sure the AR Server license is valid before retrying the upgrade

Mid-Tier Container, Smart apps & D2p rollback

Use this option if the upgrade fails before the “Disabling Rollback” stage in the Helix full stack upgrade pipeline:



This will revert the following in this sequence:

- Smart App (DWP, DWPC, IS bundles, DB components)
- ITSM DP2
- Platform DP2
- Midtier container

After this stage, a rolling update takes place, and it internally triggers metadata changes which cannot be reverted in which case please refer to the above section.

Common issues

Component	Issue Summary	Details	Commands	Resolution
Helix On Prem pipeline deployment	Upgrade fails on Set Operating mode setup	*Connection refused (Connection refused) platform-upgrade-ext:46262* *ERROR: ARException ERROR (90): Cannot establish a network connection to the AR System server; Connection refused (Connection refused) platform-upgrade-ext:46262*		Abort and rebuild the failed run of the the **HELIX_FULL_STACK_UPGRADE** pipeline and tick the **RERUN** box (**leave other boxes as they are)**
Helix On Prem pipeline deployment	Fails on Catalog deploy step	*ERROR com.bmc.arsys.rx.deployment.common.BundleOperationHandler - User [C2CDemo] is not authenticated. javax.ws.rs.ProcessingException: java.net.ConnectException: Connection refused atorg.glassfish.jersey.client.internal.HttpUrlConnector.apply(HttpUrlConnector.java:260)*		Abort and rebuild the failed run of the the **HELIX_FULL_STACK_UPGRADE** pipeline and tick the **RERUN** box (**leave other boxes as they are)**
IS URLs not loading	IS URLs like MidTier, ITSM etc., not loading https://is-poc.training.local/arsys	Browser cannot load page	#nslookup of mid-tier hostname **ssh deploymentengine** **$nslookup <mid-tier-host>** Example **$nslookup is-poc.training.local** Above should correctly resolve to load balancer IP address	1) Update DNS as required **ssh worker10** $vim /var/named/training.local.db is-poc IN A 192.168.100.31 is-poc-smartit IN A 192.168.100.31 is-poc-sr IN A 192.168.100.31 is-poc-is IN A 192.168.100.31 is-poc-restapi IN A 192.168.100.31

			Example - 192.168.100.31 for BDC lab	is-poc-atws IN A 192.168.100.31 is-poc-dwp IN A 192.168.100.31 is-poc-dwpcatalog IN A 192.168.100.31 is-poc-vchat IN A 192.168.100.31 is-poc-chat IN A 192.168.100.31 is-poc-supportassisttool IN A 192.168.100.31 **systemctl restart named** 2) Reload URL page in browser
Pipeline failure	HELIX_ITSM_INTEROPS fails as image cannot be pulled from DTR	Jenkins console output for pipeline shows below error - Verifying Pod :tctlgettenant-vpmh6 -- 0/1^ImagePullBackOff^11m Sleeping for 10s..	# Confirm required images are available in DTR and recreate DTR secret for ADE with correct credentials **$ kubectl delete secret bmc-dtrhub -n helixade** **$kubectl create secret docker-registry bmc-dtrhub --docker-server=docker.trybmc.com --docker-username=<username> --docker-password=<pas**	1) Recreate DTR secret as shown in commands section 2) Rebuild HELIX_ONPREM_DEPLOYMENT by checking only HELIX_ITSM_INTEROPS in product-deploy section **NOTE**: HELIX_ITSM_INTEROPS does not have RERUN option 3) Once HELIX_ITSM_INTEROPS completes successfully, uncheck HELIX_ITSM_INT

			sword> -n helixade **NOTE**: DTR secret for ITSM images is configured in IMAGESECRET_NAME (Innovation Suite namespace) and is different from above DTR secret "bmc-dtrhub" used in ADE namespace for pulling ADE images	EROPS in PRODUCT-DEPLOY section and check next relevant pipelines like post configuration etc.,
Pipeline failure	HELIX_ITSM_INTEROPS fails as tenant cannot be updated	Jenkins console output for pipeline shows below error Executing Command: /home/git/INTEROPS/HELIX_INTEROPS/is/task/validatePods.sh helixade tctlupdatetenant 1 JOB NAMESPACE=helixade POD_NAME=tctlupdatetenant PODNUM=1 OPERATION=JOB TIMEOUT=10800 Getting Pod List Number of Pods: 1 0/1^ContainerCreating^0s Verifying Pod :tctlupdatetenant-m2dpp Sleeping for 10s.. Getting Pod List Number of Pods: 2 Verifying Pod :tctlupdatetenant-m2dpp 0/1^Error 10s Pod: tctlupdate tenant-m2dpp is in error State: Error	NA	1) Check HELIX_PLATFORM_CUSTOMER_NAME parameter is correct in HELIX_ONPREM_DEPLOYMENT Example – adelab 2) Rebuild HELIX_ONPREM_DEPLOYMENT by checking only HELIX_ITSM_INTEROPS in product-deploy section NOTE: HELIX_ITSM_INTEROPS does not have RERUN option 3) Once HELIX_ITSM_INT

				EROPS completes successfully, uncheck HELIX_ITSM_INTEROPS in PRODUCT-DEPLOY section and check next relevant pipelines like post configuration etc.,
Pipeline failure	HELIX_SMARTAPPS_DEPLOY fails due to code checkout issue	Jenkins console output for pipeline shows below error ASK [get_helm_charts_smartapps: fatal: [deploymentengine.training.local]: cmd: /usr/bin/**** checkout --force 2021306.1.06.00 msg: Failed to checkout 2021306.1.06.00 rc: 1 stderr: |- error: pathspec '2021306.1.06.00' did not match any file(s) known to ****. stderr_lines: <omitted> stdout: " stdout_lines: <omitted> Checkout Helm charts from repo] **** FAILED! => changed=false	NA	1) Check SMARTAPPS_HELM_REPO parameter is correct as documented in HELIX_ONPREM_DEPLOYMENT 2) Rebuild HELIX_ONPREM_DEPLOYMENT by checking HELIX_GENERATE_CONFIG and HELIX_SMARTAPPS_DEPLOY in product-deploy section 3) Once HELIX_SMARTAPPS_DEPLOY completes successfully, uncheck HELIX_GENERATE_CONFIG and HELIX_SMARTAPPS_DEPLOY in PRODUCT-DEPLOY section of HELIX_ONPREM_DEPLOYMENT

				and check next relevant pipelines to continue deployment
Pipeline failure	HELIX_PLATFORM_DEPLOY pipeline fails due to AR FTS pod does not starting up externa	AR FTS pod log shows below error INFO - <TrID: INFO - <TrID: INFO - <TrID: INFO - <TrID: INFO - <TrID: java.base/sun.security.ssl.TrustStore java.base/sun.security.ssl.TrustStore java.base/sun.security.ssl.TrustStore Manager.getTrusted Certs(TrustStore Manager.java:55) INFO - <TrID: INFO - <TrID: INFO - <TrID: > Caused by: java.io.IOException: Invalid keystore format > at java.base/sun.security.provider.JavaKeyStore.engineLoad(JavaKeyStore.java:670) > at java.base/sun.security.util.KeyStoreDelegator.engineLoad(KeyStoreDelegator.java:222) > at java.base/java.security.KeyStore.load(KeyStore.java:1479) > at Manager $TrustAnchorManager.loadKeyStore(TrustStore Manager.java:365) > at INFO - <TrID: Manager $TrustAnchorManager.getTrustedCerts(TrustStore Manager.java:313) > at > at java.base/sun.security.ssl.TrustManagerFactoryImpl.enginelnit(TrustManagerFactoryImpl.java:49) 143 common frames omitted >	#CHECK LOGS **$ssh deploymentengine** **$kubectl get pods -n <is-namespace>** #exec into the platform-fts pod **$kubectl exec -it platform-fts-0 -c platform -n <is-namespace> sh** **$cd /opt/bmc/ARSystem/db** #Below log shows error **$ vi ardebug.log** #RECREATE CACERTS TO RESOLVE ISSUE **#make sure to import self-signed certificate into out of box cacerts and recreate cacerts secret** **$cp /root/class_kit/custom_cacert.pem**	1) Recreate cacerts secret with correct self-signed certificate 2) Restart platform FTS pod by deleting pod and pod starts successfully 3) Check RERUN option in HELIX_PLATFORM_DEPLOY pipeline and click Rebuild to complete execution of the pipeline 4) Uncheck above pipeline from HELIX_ONPREM_DEPLOYMENT and check next pipelines to continue deployment

			/root/cert/custom_cacert.pem **$cp /root/class_kit_is/cacerts /root/cert/cacerts** #Import custom CA certificate to out of box ITSM cacerts using below **$keytool -importcert -v -alias dsmlab -file /root/cert/custom_cacert.pem -keystore /root/cert/cacerts** **#delete cacerts secret** **$kubectl delete secret cacerts -n <is-namespace>** **#create cacerts secret** **$kubectl create secret -n <is-namespace> generic cacerts --from-file=/root/cert/cacerts --dry-run=client -o yaml | kubectl apply -f –** **#restart FTS platform pod**	

			$kubectl delete pod platform-fts-0 -n <is-namespace>	
Pod does not start	Platform FTS pod does not start due to Elastic connectivity	Platform pod is not starting up	**$ssh deploymentengine** **$kubectl get pods -n <is-namespace>** #exec into the platform-fts pod **$kubectl exec -it platform-fts-0 -c platform -n <is-namespace> sh** # check connectivity **curl --insecure https://elasticsearch-logs-opendistro-es-data-svc.helixade:9200 -u admin:admin -k**	Fix any connectivity issue from platform FTS pod to Elastic endpoint and pod should start correctly

Addendum

BMC Reference Material and Videos

Product	Description
BHOM (Webinar)	Planning for On-Prem Helix ITOM - (WebEx)
Dashboards (Webinar)	BMC Helix Dashboards Where to Start
Dashboards (Webinar)	BMC Helix Dashboards Advanced Dashboards - Part 1
Dashboards (Webinar)	BMC Helix Dashboards Advanced Dashboards - Part 2
BHCO	BHCO Migrating to 21.3 SaaS Helix Platform (WebEx)
ITSM (Webinar)	Planning for IT Service Management (ITSM) OnPrem Deployment
BMC Containerization Community Channel	Join Here

Note: The reason for showing two versions is that SaaS is made available first, followed by the On-Prem release.

1.1.2 Troubleshooting Innovation Suite On-Prem Deployments

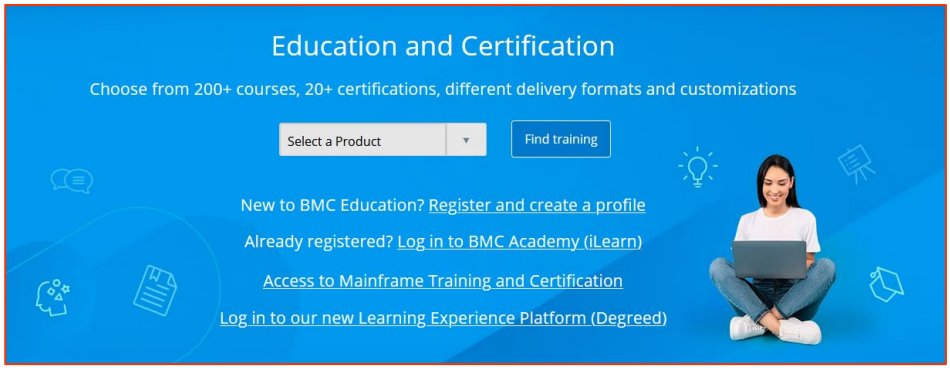
BMC Support has created several Troubleshooting Guides. Please check the documentation.

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Troubleshooting/

e) BMC Educational Materials

BMC has a comprehensive Education and Certification. Choose from 200+ courses, 20+ certifications, different delivery formats and customizations:

https://www.bmc.com/education/training-and-certification.html



f) Quick-Learn Overviews Videos for Helix BHOM, AIOps & Dashboards

There are some complementary Videos and Recording available which can be utilised to benefit the knowledge of the Helix products

Area	Description	Duration
BHOM	What's new in BMC Helix Operations Management 22.2	4 minutes
BHOM	What's new in BMC Helix Operations Management 22.1	2 minutes
BHOM	Creating an advanced enrichment policy | BMC Helix Operations Management	5 minutes
BHOM	How to create a dynamic enrichment policy in BMC Helix Operations Management 22.2	6 minutes
BHOM	How to create alarm policies | BMC Helix Operations Management	6 minutes
BHOM	How to create Event Policies | BMC Helix Operations Management	4 minutes
BHOM	How to create variate Policies | BMC Helix Operations Management	5 minutes
BHOM	Quickly start OS monitoring | BMC Helix Operations Management	4 minutes
BHOM	Sending third-party events to BMC Helix Operations Management	6 minutes
BHOM	BMC Helix Service Monitoring Overview	5 minutes
AIOps	Benefits of using BMC Helix AIOps	2 minutes
AIOps	Probable cause analysis of incidents by using BMC Helix Service Monitoring (Powered by AIOps)	2 minutes
AIOps	BMC Helix Operations Management Trial: Service Aware Monitoring	1 minute
AIOps	BMC Helix Operations Management Trial Event Noise Reduction	1 minute
AIOps	BMC Helix Operations Management Trial: Root Cause Isolation with Intelligent Automation	1 minute
Dashboards	BMC Helix Dashboards Webinar	70 minutes

BMC delivers software solutions that help IT transform digital enterprises for the ultimate competitive business advantage. We have worked with thousands of leading companies to create and deliver powerful IT management services. From mainframe to cloud to mobile, we pair high-speed digital innovation with robust IT industrialization—allowing our customers to provide amazing user experiences with optimized IT performance, cost, compliance, and productivity. We believe that technology is the heart of every business, and that IT drives business to the digital age.

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